

Answer: A

20 $P_r = I^2 r$

$$P_{total} = I^2 R + I^2 r$$

$$\text{Fraction of total power delivered to lamp} = \frac{I^2 r}{I^2 R + I^2 r} = \frac{r}{R + r}$$

Answer: A

21 Let the e.m.f. of the cell be $\mathcal{E}$ and the resistance of each lamp be R .